

UNIT – 4

Concept of cyber crime

Cyber crime is a general term describing the myriad of criminal activities carried out using a computer, network, or another set of digital devices. Consider cyber crime the umbrella over the vast range of illegal activities that cyber criminals commit. These include [hacking](#), phishing, [identity theft](#), ransomware, and malware attacks, among many others.

The reach of cybercrime knows no physical boundaries. Criminals, victims, and technical infrastructure span worldwide. With the use of technology to exploit security vulnerabilities on both a personal and enterprise level, cybercrime takes many shapes and continuously evolves. In turn, the ability to effectively investigate, prosecute, and prevent cybercrimes is an ongoing fight with many dynamic challenges.

Cybercrime poses a serious threat to individuals, businesses, and government entities and can result in significant financial loss, damaged reputation, and compromised records. As technology advances and more people rely on digital devices and networks for standard operations, the threat of cyber crime continues to increase, making it more critical than ever to take steps to protect against it.

Cyber Crime Types and Examples

1. Email Scams

Misleading schemes that take many forms. Fake emails mislead recipients, while social engineering techniques deceive people into divulging information, such as credit card numbers, or transferring money to the attacker. Phishing schemes, whereby scammers mimic legitimate brands, are a common form of [email scams](#).

2. Social Media Fraud

Scams that use social media platforms like Facebook, Twitter, Instagram, and TikTok to deceive and defraud victims. Examples include fictitious online stores, [catfishing](#), social engineering attacks, or impersonation scams. [Social media frauds](#) often exploit user trust, naivety, and a tendency to overshare personal information online.

3. Banking Fraud

Fraudulent activities that target financial institutions or their customers and stakeholders. Banking frauds most commonly result in significant financial loss or identity theft, and attacker strategies often involve sophisticated hacking and [social engineering tactics](#). Examples include credit card fraud, ATM skimming, and online banking scams.

4. eCommerce Fraud

Elaborate consumer scams that exploit weaknesses and pitfalls of online shopping technologies, like artificial or fabricated online stores, fake seller accounts, or credit card information theft. Cases of eCommerce fraud typically result in financial losses on behalf of both consumers and online retailers.

5. Malware

A highly-prevalent software attack programmed to damage and manipulate computer systems by introducing viruses, trojans, or [spyware](#) into the system. [Malware](#) is a frequent problem across many cases because it targets both individual PCs and enterprise-level computer networks. It's most commonly used for disrupting networks and stealing data from users.

6. Data Breaches

Unauthorised access or leaks of sensitive data, such as confidential information, critical records, or financial access. [Data breaches](#) can be attributed to a wide array of risk factors, such as weak passwords and cybersecurity protocols, software system vulnerabilities, or insider threats. The consequences can result in compromised data, financial damages, or tarnished reputations. [Verizon's](#) data breach investigations report highlighted that 82% of breaches involved a human element.

7. Identity Theft

In a digital context, identity theft refers to acquiring someone's [private data](#) for fraudulent or malicious purposes. Target assets of identity theft include social security numbers, date of birth, credit card details, or online accounts. Specific types include financial, medical, and tax identity theft; social media impersonation; and identity cloning, when a person uses another's identity to conceal their own.

8. Online Harassment

Involves cyberbullying, cyberstalking, and repeated acts intended to scare, harm, anger, or shame a particular individual. Today, online harassment is most prevalent on social media sites, dating apps, and forums/message boards. Examples of online harassment include sending inappropriate and unsolicited messages, making clear and intentional threats, or distributing sensitive photos or videos of a victim.

Phishing

What is phishing?

Phishing is a type of cyberattack that uses fraudulent emails, text messages, phone calls or websites to trick people into sharing sensitive data, downloading [malware](#) or otherwise exposing themselves to cybercrime.

Phishing attacks are a form of [social engineering](#). Unlike other [cyberattacks](#) that directly target networks and resources, social engineering attacks use human error, [fake stories](#) and pressure tactics to manipulate victims into unintentionally harming themselves or their organizations.

In a typical phishing scam, a [hacker](#) pretends to be someone the victim trusts, like a colleague, boss, authority figure or representative of a well-known brand. The hacker sends a message directing the victim to pay an invoice, open an attachment, click a link or take some other action.

Because they trust the supposed source of the message, the user follows the instructions and falls right into the scammer's trap. That "invoice" might lead directly to a hacker's account. That attachment might install [ransomware](#) on the user's device. That link might take the user to a website that steals credit card numbers, bank account numbers, login credentials or other [personal data](#).

Cyber Crime Prevention

- **Multifactor Authentication**

[Multifactor authentication](#) (MFA), commonly used as two-factor authentication, is a common security protocol that prevents data breaches, hacks, and other direct cyber-attacks. In simple terms, this process requires users to provide two or more forms of identification to authenticate access to their accounts, such as needing a password and an access code sent to a device. Now a best practice protocol for organisations, MFA adds extra layers of cybersecurity to online accounts, making it much more challenging for attackers to access your data.

- **Virtual Private Networks**

A Virtual Private Network (VPN) is a service that enables users to browse the Internet with reinforced security and anonymity. [VPNs](#) are engineered to encrypt online activity, making it far more difficult for cyber-attackers to intercept and steal your data. VPNs act as intermediaries between your device and the targeted server, adding their own encryption layer and routing communication via their own servers. VPNs are especially effective in helping protect against email frauds like phishing scams by masking your IP address and location.

- **Password Managers**

Cyber criminals frequently attack password credentials. In addition to creating secure, difficult-to-hack passwords, password managers are software applications that securely store multiple login credentials in an encrypted database, all of which are locked behind a master password. Password managers are commonly used by

organisations, remote teams, and individuals to provide extra security protection when surfing the web while safely maintaining passwords in a safe space. The most common password managers include 1Password, KeePass, LastPass, and Apple's iCloud Keychain. However, some [password managers](#) come with risks.

- **Data Backup and Recovery**

Many forms of cyber-attacks can result in critical data loss, which can have severe financial and operational repercussions for both individuals and organisations. Data backup and recovery solutions can help mitigate the damages from data loss by creating backup copies of data and ensuring a faster recovery in the event of a ransomware attack, data breach, or another form of cyber-attack. Regularly [archiving data](#) is an essential security protocol to ensure that you can recover your data in the event of an attack.

- **Email Security Solutions**

Email accounts are one of the most frequently exploited channels for cyber-attackers to breach access to sensitive data and private information. Specialised email security technologies can be leveraged to prevent this activity, which includes solutions like email encryption, spam filters, and antivirus software. [Encryption](#) is a powerful technology that protects email content from interception. Spam filters detect and prevent unwarranted and malicious emails from reaching your inbox, while antivirus software detects and removes malicious attachments from emails.

Case Study

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Security threats to e-commerce payment system

In today's digital age, electronic payment systems have become ubiquitous, offering convenience and efficiency to businesses and consumers alike. Among these systems, [electronic checks](#) (e-checks) stand out as a popular method for transferring funds electronically. However, with the convenience comes the need for robust security measures to safeguard transactions and protect sensitive financial information from malicious actors. In this comprehensive guide, we delve into the world of e-checks and explore the essential security measures necessary to ensure the integrity and safety of electronic payment systems.

- **Fraudulent Activities:**

Malicious actors may attempt to exploit vulnerabilities in e-check systems to perpetrate fraudulent activities, such as unauthorized transactions, account takeovers, and counterfeit checks. These fraudulent activities can result in significant financial losses and damage to the reputation of the affected parties.

- **Phishing and Social Engineering:**

Phishing attacks targeting users of electronic payment systems are a common tactic employed by cybercriminals. By impersonating legitimate entities or using deceptive tactics, attackers attempt to trick individuals into divulging sensitive information such as login credentials or banking details.

- **Malware and Ransomware:**

Malicious software, including malware and ransomware, pose a significant threat to e-check systems. These malicious programs can infiltrate systems, compromise security controls, and disrupt operations, leading to financial losses and operational downtime.

Digital Signature

A **digital signature** is a cryptographic technique used to verify the **authenticity, integrity,** and **non-repudiation** of a digital message or document. It ensures that the message was created by a known sender and that it has not been altered during transmission.

Key Components of Digital Signature

1. Key Generation Algorithm

Digital signatures use **asymmetric cryptography**, which involves a pair of keys:

- **Private Key:** Kept secret by the owner and used to create the signature.
- **Public Key:** Shared with others and used to verify the signature.

This key pair ensures secure authentication during digital transactions.

2. Signing Algorithm

To create a digital signature:

1. A **hash function** is applied to the original message to generate a fixed-length value called a **message digest**.
2. This message digest is then **encrypted using the sender's private key**.
3. The encrypted hash value forms the **digital signature**.

Instead of encrypting the entire message, only the hash is encrypted because:

- Hash values are much shorter
- Hashing is faster than encryption
- It improves efficiency without reducing security

3. Signature Verification Algorithm

At the receiver's side:

1. The digital signature is **decrypted using the sender's public key**, producing the original message digest.
2. The receiver independently computes the hash of the received message using the same hash function.
3. Both hash values are compared:
 - If they match → signature is **valid**
 - If they differ → message integrity is compromised

